

Central Asian University

INTERNSHIP DIARY

Student Name:	Ismoil Rahimov
Student ID:	230514
Programme:	Computer Science / IT
Company:	Alif Transportation LLC
Department:	IT & Data Operations
Duration:	01 March 2026 – 30 April 2026 (8 weeks)

Week 1 (01–07 March 2026)

Theme: Orientation & Environment Setup

Mon 01 Mar	Arrived at the company. Met the IT team and supervisor. Received company laptop and access credentials. Overview of company operations: fleet management, driver data, dispatch.
Tue 02 Mar	Installed required software: Python 3.10, VS Code, OpenCV 4.8, Git. Cloned the internal data scripts repository. Read company data-handling policies and signed NDA.
Wed 03 Mar	Introduction to the company's existing image processing tasks: document scanning for driver licenses, barcode reading for vehicle tags. Reviewed OpenCV documentation.
Thu 04 Mar	Set up GitHub repository for the internship project. Created project folder structure as per university guidelines. Wrote initial README draft.
Fri 05 Mar	Team meeting: discussed project scope. Decided to focus on QR code detection for driver document verification. Prepared weekly summary report for supervisor.

Week 2 (08–14 March 2026)

Theme: Dataset Collection & Literature Review

Mon 08 Mar	Began collecting test images of QR codes from company documents and product barcodes in the logistics warehouse.
Tue 09 Mar	Generated additional QR codes using qr-code-generator.com to ensure coverage of various data types (URL, text, phone number).
Wed 10 Mar	Read three research papers on QR code detection algorithms. Took notes for the literature review section of the technical report.
Thu 11 Mar	Organised 50 test images into three categories: ideal, rotated, and real-world. Annotated each image with expected output.
Fri 14 Mar	Presented dataset plan to supervisor. Received feedback to add more varied lighting conditions. Updated dataset accordingly.

Week 3 (15–21 March 2026)

Theme: Core Implementation — QR Detection Module

Mon 15 Mar	Started implementing the main detection function using cv2.QRCodeDetector().detectAndDecodeMulti(). Tested on 5 sample images.
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Tue 16 Mar	Added <code>draw_detection()</code> function to annotate frames with green polygons and decoded text overlays.
Wed 17 Mar	Implemented <code>process_image()</code> function to handle single image input. Results saved automatically to <code>results/</code> directory.
Thu 18 Mar	Debugging session: fixed issue with empty decoded strings causing index errors. Added data validation check.
Fri 21 Mar	Code review with supervisor. Improved variable naming and added docstrings. Committed clean version to GitHub.

Week 4 (22–28 March 2026)

Theme: Real-Time Webcam Module & Testing

Mon 22 Mar	Implemented <code>run_webcam()</code> function with <code>cv2.VideoCapture()</code> . Added real-time FPS counter display.
Tue 23 Mar	Tested webcam detection with various QR codes displayed on phone screen at different distances and angles.
Wed 24 Mar	Performance profiling: system achieves 27 FPS on office laptop. Identified bottleneck in <code>imshow()</code> call — acceptable for demo purposes.
Thu 25 Mar	Implemented <code>evaluate_dataset()</code> function for batch testing. Ran evaluation on all 50 test images and logged results.
Fri 28 Mar	Wrote interim progress report for university supervisor. Uploaded all code to GitHub with proper commit messages.

Week 5 (29 Mar – 04 Apr 2026)

Theme: Evaluation & Metrics

Mon 29 Mar	Analysed evaluation results: 47/50 images detected correctly (94%). Investigated 3 failure cases — all due to severe blur or occlusion.
Tue 30 Mar	Calculated evaluation metrics: detection rate, false positive rate, average decode time per image (~36 ms).
Wed 01 Apr	Created result visualisations: annotated sample images, accuracy bar chart by category.
Thu 02 Apr	Discussed results with supervisor. Noted that <code>WeChatQRCode</code> model could improve accuracy on hard cases.
Fri 04 Apr	Updated GitHub README with results table and usage instructions.

Week 6 (05–11 April 2026)

Theme: Report Writing

Mon 05 Apr	Started writing the Technical Report. Completed Abstract, Introduction, and Literature Review sections.
Tue 06 Apr	Wrote Problem Statement, Objectives, and Dataset Description. Added figures and tables.
Wed 07 Apr	Wrote Methodology section with pipeline diagram description. Included code snippets with explanations.
Thu 08 Apr	Completed Results & Discussion and Evaluation Metrics sections. Proofread the full report.
Fri 11 Apr	Submitted draft report to company supervisor for review. Received minor correction feedback.

Week 7 (12–18 April 2026)

Theme: Presentation Preparation

Mon 12 Apr	Applied supervisor corrections to technical report. Final report compiled as PDF.
Tue 13 Apr	Created presentation slides covering all report sections. Used professional design template.
Wed 14 Apr	Rehearsed 12-minute presentation with timer. Adjusted slide content based on timing feedback.
Thu 15 Apr	Practice presentation in front of IT team. Received constructive feedback on demo section.
Fri 18 Apr	Polished presentation based on feedback. Uploaded final version to GitHub.

Week 8 (19–30 April 2026)

Theme: Final Submission & Wrap-Up

Mon 19 Apr	Final code cleanup and documentation review. Ensured all functions have complete docstrings.
Tue 20 Apr	Organised GitHub repository: confirmed all folders present (src/, dataset/, results/, report/, presentation/).
Thu 24 Apr	Wrote internship diary and internship report. Compiled all deliverables.
Fri 25 Apr	Exit interview with company supervisor. Received positive performance evaluation.
Wed 30 Apr	Submitted all university deliverables. Internship officially completed.